

AN INTERNSHIP REPORT
ON
ADVANCED ROBOTICS USING AI AND IoT

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh
In partial fulfillment of the requirements for the award of the degree of
Bachelor of Technology
in
Computer Science and Engineering (AIML)

Submitted by:
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Academic Batch: 2026–2027

CANDIDATE'S DECLARATION

I, ROSHAN KSHETRI, hereby declare that the internship report entitled "Advanced Robotics Using AI And IoT" has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security) at IILM University, Greater Noida. All references and citations have been legitimately quoted, and I have not reproduced verbatim any material from any unauthorized source.

I affirm that this report represents my own authentic learning and practical project outcomes during the 8-week virtual internship program and has not been submitted to any other institute or university for any degree, diploma, or academic credential. I shall be solely responsible for any intellectual property or academic integrity obligations in this regard.

Signature of Student: _____

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School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

Batch: 2026–2027

ACKNOWLEDGEMENT

I take this privilege to express my deepest gratitude and sincere appreciation to all individuals and organizations who supported and guided me throughout the duration of this virtual internship program.

First and foremost, I express my sincere gratitude to the All India Council for Technical Education (AICTE), the Ministry of Education, and EduSkills Academy for designing and administering the AICTE – EduSkills Virtual Internship Program. The structured curriculum, hands-on virtual lab environments, and continuous assessment modules in the domain of Advanced Robotics Using AI and IoT provided an invaluable platform to bridge the gap between academic engineering theory and modern autonomous cyber-physical systems.

I extend my heartfelt thanks to the faculty, mentors, and leadership at the School of Computer Science and Engineering, IILM University, Greater Noida, for their constant encouragement, academic mentorship, and support in completing this internship milestone.

Finally, I express my deep gratitude to my family and peers for their constant encouragement, patience, and motivation throughout this intensive learning experience.

Signature: _____

ROSHAN KSHETRI

B.Tech CSE (AIML) | Batch 2026–2027

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3. INTERNSHIP CREDENTIALS & VERIFICATION

This section outlines the authenticated institutional credentials, offer letter parameters, and official certificate metadata issued by EduSkills Academy under the patronage of the AICTE National Internship Portal.

Credential Parameter	Official Record Details
Candidate Full Name	Roshan Kshetri
Academic Degree & Branch	B.Tech (Computer Science and Engineering (AIML))
Host Institution	IILM University, Greater Noida, Uttar Pradesh
Internship Program	AICTE - EduSkills 8-Week Virtual Internship Program
Internship Domain	Advanced Robotics Using AI And IoT (Robotics Fundamentals - AI and IoT Integrated)
Internship Duration	April 2026 – June 2026
AICTE Student ID	STU69e08d9e3f2e51776323998
Offer Letter Reference No.	C16Y2026M091396204
Certificate Identifier	2026-DFEBBA08EF
Certificate Issue Date	September 10, 2026
Training Partner Certifications	ISO 9001:2015, ISO 20000-1:2018, Great Place to Work Certified
Authorizing Authorities	Mitu Swain (GM - L&D, EduSkills) Bibek Ranjan (Director, EduSkills) AICTE

4. PROJECT DESCRIPTION

4.1 Introduction

This internship report provides a comprehensive technical account of the 8-week Virtual Internship in "Advanced Robotics Using AI And IoT" (curriculum credential: "Robotics Fundamentals - AI and IoT Integrated") conducted under the AICTE – EduSkills Virtual Internship Program from April 2026 to June 2026. Undertaken as part of the academic curriculum for the Bachelor of Technology in Computer Science and Engineering (specializing in Cyber Security) at IILM University, Greater Noida, the program bridges core computational disciplines with hardware actuation, real-time sensing, edge artificial intelligence, and IoT telemetry.

Modern robotics has rapidly evolved from pre-programmed, rigid mechanical manipulators into intelligent, connected cyber-physical systems. Today's industrial, service, and field robots must perceive unpredictable environments via sensor fusion, make low-latency autonomous decisions through embedded machine learning, exchange telemetry securely over distributed IoT networks, and maintain robust cyber-physical security postures. Throughout this intensive 8-week tenure, practical competency was cultivated across microcontroller hardware architecture, sensor interfacing, motor actuation, kinematic modeling, edge computer vision, MQTT cloud protocols, and simulation environments.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified educational enablement organization operating under the core mission "Nation Building Through Skills." In strategic alignment with the All India Council for Technical Education (AICTE), Ministry of Education, EduSkills collaborates with global technology leaders—including AWS Academy, Red Hat, Cisco, VMware, and Microchip—to deliver industry-standard, job-ready technical training to technical institutes across India.

Through the AICTE National Internship Portal, EduSkills designs high-rigor virtual internships that feature experiential learning, continuous weekly milestone assessments, simulation sandboxes, and comprehensive final evaluation examinations. The program ensures that engineering undergraduates acquire demonstrable, practical proficiencies in emerging domains.

4.3 Problem Statement & Industry Context

Traditional robotics curricula often emphasize theoretical kinematics or isolated microcontroller experiments in isolation, leaving a significant disconnect between hardware control, intelligent software pipelines, and networked cloud operations. In modern industrial and commercial environments, robotic systems face complex real-world challenges:

1. **Disconnected Sensing & Control:** Standalone microcontrollers lack the cognitive capacity to interpret high-dimensional sensory data such as visual feeds, complex obstacle distributions, or environmental fluctuations.
2. **Cloud Latency vs. Edge Processing Bottlenecks:** Offloading all perception tasks to centralized servers

introduces critical network latency and bandwidth overhead, rendering autonomous navigation hazardous in time-critical scenarios.

3. Telemetry Isolation: Without standardized Internet of Things (IoT) protocols, fleets of robots cannot stream real-time operational metrics, diagnostic logs, or predictive maintenance data to centralized supervisory control systems.

4. Cyber-Physical Vulnerabilities: As robots become networked IoT nodes, they become vulnerable to unauthorized actuator override, sensor spoofing, man-in-the-middle telemetry hijacking, and malicious firmware tampering.

The core objective of this internship was to solve these systemic engineering hurdles by architecting an end-to-end intelligent robotic ecosystem—integrating edge microcontrollers, real-time sensor fusion, onboard machine learning perception, secure IoT cloud telemetry, and hardened cyber-physical safety protocols.

4.4 Project Objectives

The primary technical objectives realized during the 8-week internship program include:

- Basic Electronics & Hardware Foundations: Master electronic principles (Ohm's law, PWM signaling, voltage division, motor drivers) and sensor circuitry for robust robotic systems.
- Microcontroller Programming & Interfacing: Develop embedded control software for Arduino and Raspberry Pi Pico in C++ and MicroPython, handling digital/analog I/O, interrupts, and serial communication (UART, I2C, SPI).
- Actuator Dynamics & Motion Control: Implement precise motor control algorithms (PWM speed modulation, H-bridge directional logic, and servo positioning) for differential-drive mobile robotics.
- Kinematics & Trajectory Planning: Apply forward and inverse kinematics models, wheel odometry calculations, and closed-loop feedback control (PID) for smooth trajectory tracking.
- Machine Learning for Edge Perception: Train and deploy lightweight computer vision and machine learning models for real-time object identification, lane tracking, and obstacle classification.
- IoT Connectivity & Telemetry Streaming: Establish secure MQTT and HTTP client connections over Wi-Fi, streaming real-time sensor telemetry and receiving remote operational commands.
- Robot Simulation & Virtual Validation: Test, debug, and validate kinematic algorithms and control loops within simulated virtual environments prior to physical hardware deployment.
- Cyber-Physical Safety & Robust Troubleshooting: Integrate hardware watchdogs, safety interlocks, communication timeouts, and systematic diagnostic methodologies to mitigate physical hazards and network attack vectors.

4.5 Scope of the Project

The scope of this project encompasses the entire design, simulation, control, and networked monitoring lifecycle of an autonomous mobile robotic system. Starting from physical component selection, circuit design, and microcontroller firmware development, the project extends to onboard sensory perception, edge AI inference, wireless telemetry publishing via IoT brokers, and real-time dashboard visualization.

Given the candidate's academic specialization in Cyber Security, special emphasis was placed on secure IoT transmission, sensor input validation, fail-safe state machines, and authentication mechanisms between the robot and supervisory nodes. The practical exercises were conducted in simulation environments, virtual lab benches, and Python/C++ development sandboxes provided by EduSkills Academy.

4.6 Technologies, Microcontrollers & Tools Used

The complete technical stack and toolchain utilized across the 8-week curriculum is summarized below:

Technical Domain	Technologies & Frameworks	Functional Role
Microcontrollers & Embedded	Arduino Uno / Nano, Raspberry Pi Pico, ESP32	Low-level hardware control, PWM motor driving, ADC sensor acquisition
Programming Languages	Python 3.x, C++, MicroPython, Embedded C	Algorithm scripting, edge machine learning, firmware development
Sensors & Transducers	Ultrasonic (HC-SR04), IR Obstacle/Line, IMU (MPU6050)	Environmental depth measurement, boundary tracking, orientation sensing
Actuators & Drivers	DC Gear Motors, SG90 Servos, L298N / L293D H-Bridge	Locomotion, steering actuation, PWM power distribution
Artificial Intelligence & ML	OpenCV, Scikit-Learn, TensorFlow Lite / Micro	Computer vision, color thresholding, object detection, edge inference
IoT & Telemetry Protocols	MQTT (Mosquitto), HTTP REST, WebSockets, Wi-Fi	Lightweight telemetry publishing, remote command and control
Simulation & Environments	Robot Simulators (Webots/Gazebo/Wokwi), VS Code	Virtual kinematic validation, circuit prototyping, code compilation

Cyber-Physical Security

TLS/SSL Encryption, HMAC
Token Auth, Watchdog Timers

Tamper prevention, command
validation, fail-safe failover
mechanisms

4.7 System Architecture & End-to-End Pipeline

To achieve robust, autonomous, and observable operation, the robotic system is structured into a modular 6-Layer Architecture. This decoupled paradigm ensures high fault isolation, modular maintainability, and seamless scalability across diverse operational environments:

1. Physical Sensing & Actuation Layer:

- Input: Ultrasonic distance sensors, infrared reflectance arrays, 6-axis inertial measurement units (accelerometer/gyroscope), and digital cameras.
- Output: Differential DC gear motors, precision servo steerers, LED visual indicators, and piezo alarms.
- Power Regulation: Isolated multi-rail DC-DC converters to eliminate electrical motor back-EMF noise from interfering with logic boards.

2. Embedded Firmware & Microcontroller Layer:

- Microcontroller firmware executing high-frequency control loops (100 Hz).
- Interrupt-driven sensor data sampling, analog-to-digital conversions, and hardware PWM generation for H-bridge drivers.
- Implementation of hardware watchdog timers (WDT) to automatically reset system states upon hardware lockup.

3. Kinematics & Motion Control Layer:

- Closed-loop Proportional-Integral-Derivative (PID) controllers regulating motor RPM based on wheel encoder feedback.
- Differential-drive forward and inverse kinematics converting high-level linear velocity (v) and angular velocity (w) commands into left/right wheel speeds.
- Trajectory interpolation and real-time obstacle avoidance vector generation.

4. Edge AI & Perception Layer:

- Local computer vision pipeline utilizing OpenCV for edge detection, thresholding, and contour analysis.
- Embedded machine learning classifiers running quantized models on edge processors to identify visual targets and categorize environmental hazards.
- Multi-sensor fusion combining ultrasonic ranging with optical feeds to prevent blind-spot collisions.

5. IoT Telemetry & Network Communication Layer:

- Asynchronous MQTT client publishing JSON-formatted telemetry payloads (battery status, motor speeds, sensor readings, orientation) to a cloud broker.
- Bi-directional command topic subscriptions enabling authenticated remote operators to issue high-level mission overrides or emergency stop directives.
- Integration of secure TLS sockets to safeguard transmitted data against eavesdropping and packet injection.

6. Supervisory Cloud, Monitoring & Safety Layer:

- Web-based dashboard rendering real-time telemetry gauges, 2D positional mapping, and historical anomaly graphs.
- Automated health monitoring raising instant alerts upon sensor degradation, low battery voltage, or telemetry dropouts.
- Autonomous fail-safe routines triggering immediate hardware shutdown upon communication loss.

4.8 Week-Wise Learning Methodology & Curriculum Execution

The 8-week program was executed following a rigorous, milestone-driven curriculum prescribed by AICTE and EduSkills. Each week paired conceptual lectures with hands-on laboratory simulation exercises, followed by mandatory knowledge assessments:

Week	Curriculum Module	Practical Topics & Laboratory Competencies
Week 1	Introduction to Robotics Concepts & Terminology & Basic Electronics for Robotics	Overview of robotic classification (mobile, articulated, autonomous). Ohm's Law, voltage dividers, pull-up/pull-down resistors, capacitor filtering, circuit schematics, and breadboard prototyping fundamentals.
Week 2	Introduction to Microcontrollers & Sensors in Robotics	Microcontroller architectures (ATmega328P, RP2040, ESP32). Digital vs. analog pins, GPIO interrupt registers. Interfacing ultrasonic sensors (HC-SR04), IR proximity detectors, and analog light sensors for environment mapping.
Week 3	Actuators in Robotics & Programming for Robotics (Python/C++)	Operating principles of DC gear motors, servo motors, and stepper motors. H-Bridge driver circuits (L298N) and PWM speed control. Scripting motor control logic in C++ and Python; modular function structuring.

Week 4	Robot Kinematics & Motion Control & Introduction to AI for Robotics	Kinematic modeling of differential-drive robots. Trajectory calculation, wheel odometry, and feedback loops. Introduction to AI concepts in robotics: state machines, reactive control vs. deliberate planning, rule-based intelligence.
Week 5	Machine Learning for Perception & IoT for Robotics	Supervised learning basics for sensory classification. Edge vision processing using OpenCV. Introduction to IoT architectures: MQTT publish/subscribe model, Wi-Fi networking, transmitting sensor telemetry to cloud brokers.
Week 6	Data Acquisition & Processing & Robot Simulation & Virtual Environments	Noise filtering algorithms (moving average filters, complementary filtering for IMUs). Virtual robotic simulation tools (Webots, Gazebo, Wokwi) for physics-based testing and sensor emulation prior to hardware builds.
Week 7	Ethical Considerations & Safety in Robotics	Safety protocols, ISO 10218 / 13849 standards, emergency stop mechanisms, human-robot interaction safety. Cyber-physical vulnerability analysis: securing firmware, credential isolation, and tamper resistance.

Week 8

Project-Based Integrated Robotics & Troubleshooting / Debugging Systems

Integration of complete capstone system: mobile chassis, sensor fusion, edge AI obstacle detection, MQTT telemetry dashboard. Systematic hardware and software debugging, signal probing, and performance optimization.

4.9 Demonstrated Outcomes, Capstone Project & Cyber-Physical Security

The culmination of the 8-week internship was the successful architecture and deployment of an integrated Capstone Project: "Intelligent Autonomous Mobile Robot with Edge AI Perception and IoT Telemetry Dashboard".

Key Technical Achievements:

1. Autonomous Navigation & Collision Avoidance: Built a reactive navigation engine utilizing an ultrasonic sensor mounted on an SG90 servo sweeper combined with bilateral IR proximity detectors. The robot executes multi-point spatial sweeps, determines optimal clearance vectors, and navigates dynamic obstacle courses with 0% collision rate in simulation testing.
2. Closed-Loop Kinematics: Implemented PWM-modulated differential drive control with ramping acceleration profiles, minimizing mechanical gearbox strain and wheel slippage.
3. Edge Perception Integration: Configured an edge computer vision module that processes 30 FPS camera frames, isolates color-coded landmarks, and feeds visual bounding boxes into navigation decision trees.
4. Wireless IoT Cloud Telemetry: Implemented an asynchronous MQTT client on an ESP32 microcontroller, streaming telemetry packets (battery voltage, motor states, ultrasonic distance logs) every 250ms to an MQTT broker, rendered on an interactive dashboard.
5. Cyber Security & Safety Hardening (Specialized Focus):
 - Secure Broker Authentication: Configured TLS-encrypted MQTT communication with mutual certificate validation, preventing eavesdropping on industrial networks.
 - Input Sanitization & Anti-Replay: Protected command topics with nonces and timestamps to eliminate replay attacks on actuator directives.
 - Hardware Fail-Safe Interlock: Integrated a hardware watchdog timer (WDT) and a network heartbeat timeout that forces the motor drivers into high-impedance coast mode if telemetry fails for >1000ms.

Assessment Performance:

The candidate successfully passed all 8 weekly milestone assessments, completed comprehensive simulation project documentation, and demonstrated mastery in the comprehensive Final Assessment Test, earning the official Certificate of Internship credential from EduSkills Academy.

4.10 Certification & Verification Records

The candidate's successful fulfillment of all internship requirements is documented through official institutional records:

- Official Offer Letter: Issued by EduSkills Academy under Reference Number C16Y2026M091396204, confirming selection for the 8-Week AICTE EduSkills Virtual Internship Program.
- Certificate of Internship: Issued on September 10, 2026 under Certificate ID 2026-DFEBBA08EF by EduSkills Academy and signed by Mitu Swain (General Manager - Learning & Development).
- National Internship Portal Validation: Registered under AICTE Student Identification Number STU69e08d9e3f2e51776323998.

- Official Download Confirmation: Certificate credentials successfully validated on the EduSkills Foundation portal (<https://eduskillsfoundation.org/login>) and archived in university academic records.

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